Weil Conjectures

Chapter 14: Purity of real sheaves

Snapshot 2026-09-26, revision 8e4cb88e2a

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14 Purity of real sheaves

In Chapter 17* the Fourier transform of a pure sheaf produces a lisse sheaf M' whose Frobenius eigenvalues we cannot compute. What we do know is that a twist of M' is a quotient of a sheaf whose traces are real, and from this alone we must conclude that M' is pure. In Chapter 18* the same need returns for the direct images of a family of curves, which we can make real but do not know to be mixed. Both arguments rest on one theorem (Theorem 14.26):

> on a smooth geometrically connected curve, every irreducible constituent of a real lisse Weil sheaf is pure, of weight equal to its determinant weight.

Before this theorem we know the weights of one kind of sheaf only: a lisse sheaf of rank one on a curve has finite geometric monodromy, and therefore it is pure (Section 14.1). In particular the determinant of every lisse sheaf is pure, and its weight divided by the rank, the determinant weight, is the average weight of the eigenvalues at each point. The theorem says that on an irreducible constituent of a real sheaf no eigenvalue deviates from this average.

The tool is the trace formula. On an affine curve U_0 it expresses the L-function of a lisse sheaf $\mathcal{G}$ as

$$L(U_0,\mathcal{G},t)=\prod_{x\in|U_0|}\det(1-\mathrm{Frob}_x\,t^{\deg(x)}\mid\mathcal{G}_{\bar{x}})^{-1}=\frac{\det(1-Ft\mid H^1_c(U,\mathcal{G}))}{\det(1-Ft\mid H^2_c(U,\mathcal{G}))}.$$

The poles of the right-hand side are controlled by H^2_c, which is easy to compute (Lemma 14.20). On the left, a large eigenvalue at a point x gives a pole of the local factor at x close to the origin. But this pole could be cancelled by a zero of the product of the other factors, as in $(1-2t)^{-1}\cdot(1-2t)=1$. Positive factors, however, cannot cancel in this way. Indeed, suppose all factors are power series with nonnegative coefficients and constant term 1. Then the coefficients of each factor are bounded by those of the product, and Lemma 14.24 shows that no factor can converge on a smaller disc than the product.

This idea is Rankin's. In 1939 he improved the bound for Ramanujan's function $\tau(n)$ to $O(n^{29/5})$ by studying the Dirichlet series $\sum_n \tau(n)^2 n^{-s}$, whose coefficients are nonnegative. Deligne used the same argument in his first proof [Del74].

For the Euler factors to be positive, the traces must be real. We therefore call a lisse Weil sheaf real if its local characteristic polynomials have real coefficients under the fixed embedding ι of Convention `situation-weil` (Definition 14.21). For example, every sheaf with rational characteristic polynomials is real, such as a Tate twist of the constant sheaf. On an even tensor power $\mathcal{G}^{\otimes 2m}$ of a real sheaf the traces are $2m$th powers of real numbers, and this makes the Euler factors positive. From this, Proposition 14.25 bounds each local eigenvalue α of $\mathcal{G}$ by $|\iota(\alpha)|^2 \le q_x^{\beta}$, where β is the largest determinant weight of the constituents.

This is not yet the theorem, because a real sheaf need not be pure. For example, the sheaf $\overline{\mathbf{Q}}_\ell \oplus \overline{\mathbf{Q}}_\ell(-1)$ is real, with constituents of weights 0 and 2, but the bound only gives 2 for both. To separate the constituents we pass to exterior powers. In $\bigwedge^2(\overline{\mathbf{Q}}_\ell \oplus \overline{\mathbf{Q}}_\ell(-1)) = \det \overline{\mathbf{Q}}_\ell(-1) \otimes \overline{\mathbf{Q}}_\ell$ the factor $\overline{\mathbf{Q}}_\ell$ appears multiplied by the known eigenvalue q_x of $\det \overline{\mathbf{Q}}_\ell(-1)$, so the bound 2 for this summand leaves the bound 0 for $\overline{\mathbf{Q}}_\ell$ itself. In general, Proposition proposition-determinant-weights-tensor-operations finds the exterior power in which a given group of constituents carries the largest determinant weight, and the same argument applies there. Positivity, however, only ever gives upper bounds. The lower bound comes from applying the same argument to the dual, and in Chapter 19* this role is played by Poincaré duality.

14.1 Sheaves of rank one

Rankin's argument bounds the local eigenvalues of a sheaf by its determinant weights, so it needs the determinant of every lisse sheaf to be pure (Corollary 14.8). Since a determinant has rank one, we prove more generally that every rank-one lisse Weil sheaf on a smooth geometrically connected curve is pure (Corollary 14.6), and the later sections Sections 14.2–Section 14.4 are built on this fact.

In rank one the question is about characters. Let U_0 be a smooth geometrically connected curve and $s \in W(U_0)$ a lift of Frobenius of degree one (Convention `situation-weil`). Then a rank-one lisse Weil sheaf is a character $\chi : W(U_0) \to E^\times$, continuous on $\pi_1(U)$. If χ is trivial on $\pi_1(U)$, the sheaf is one of the sheaves $\mathcal{B}_b$ with $b = \chi(s)$; at a closed point x its eigenvalue is $b^{\deg(x)}$, so it is pure of weight $2\log_q|\iota(b)|$. A Kummer sheaf is the next simplest case. If $n \mid q-1$, the cover $t \mapsto t^n$ of $\mathbf{G}_m$ is a μ_n-torsor, and so a character of $\mu_n(k_0)$ gives a rank-one sheaf on $\mathbf{G}_{m,k_0}$ whose geometric monodromy is cyclic of order dividing n. Its eigenvalues are values of the character on $\mu_n(k_0)$, hence roots of unity, and it is pure of weight 0.

Up to a twist, every rank-one sheaf behaves like these two. Indeed, by Theorem 14.5 the image $\chi(\pi_1(U))$ is finite. It follows that $\chi(\mathrm{Frob}_x)$ is $\chi(s)^{\deg(x)}$ times a root of unity, so the sheaf is pure of weight $2\log_q|\iota(\chi(s))|$.

Finiteness is a statement about Weil sheaves, because on the geometric curve there are rank-one local systems with infinite monodromy. For example, the tame fundamental group of $\mathbf{G}_{m,k}$ has the quotient $\mathbf{Z}_\ell(1) \simeq \mathbf{Z}_\ell$, and $a \mapsto (1+\ell)^a$ is a character of infinite order. But it has no Weil

structure. Indeed, a Weil structure forces $\chi(s^{-1}gs) = \chi(g)$ for $g \in \pi_1(U)$, while conjugation by s acts on $\mathbf{Z}_\ell(1)$ by multiplication by a power $q^{\pm 1}$. A character of $\mathbf{Z}_\ell(1)$ that extends to the Weil group therefore satisfies $\chi(g)^{q-1} = 1$, and it has finite order.

The general proof follows the same pattern. Since there is no explicit quotient to compute with, it uses a logarithm instead: a power χ^N takes values in $1 + \mathfrak{m}_E^r$, where the ℓ-adic logarithm converges, and then $\log \chi^N$ is an additive character of $\pi_1(U)$ invariant under conjugation by s, that is, a class in $H^1(U, E)$ fixed by Frobenius (Lemma 14.4). Such a class is zero (Proposition 14.3), so χ^N is trivial on $\pi_1(U)$.

For a proper curve, Frobenius fixes no class in H^1 because the Jacobian has only finitely many rational points. If C_0 is a proper curve with Jacobian J_0, Kummer theory and Poincaré duality identify $\varprojlim_n H^1(C, \mathbf{Z}/\ell^n)$ with the dual of the Tate module $T_\ell J$ (Proposition 14.2). A fixed vector in the dual therefore means that 1 is an eigenvalue of Frobenius on $V_\ell J = T_\ell J \otimes \mathbf{Q}_\ell$, and after scaling, Frobenius would then fix a nonzero element of $T_\ell J$. Such an element is a compatible system of ℓ-power torsion points of J defined over k_0, of unbounded order. These points cannot all lie in the finite group $J_0(k_0)$, and so there is no such element (Proposition B.15*).

For an open curve, the residue sequence contains in addition the boundary term $H^0(D, E(-1))$. A power of Frobenius acts on it by a power of $q \neq 1$, so it has no fixed vector either. For example, for $\mathbf{G}_m$ this is the computation above, since there $H^1(\mathbf{G}_{m,k}, \mathbf{Q}_\ell) = \mathbf{Q}_\ell(-1)$, with Frobenius acting by q.

14.1.1 Frobenius invariants

Over $\mathbf{C}$ the link between H^1 of a curve and its Jacobian is classical. There the Jacobian is a torus V/Λ with $\Lambda = H_1(C, \mathbf{Z})$, and $H^1(C, \mathbf{Z}) = \mathrm{Hom}(\Lambda, \mathbf{Z})$. Proposition 14.2 is the ℓ-adic version, with $T_\ell J$ in place of $\Lambda \otimes \mathbf{Z}_\ell$. Here the link with the Jacobian is Kummer theory, which is a statement about μ_{ℓ^n} and line bundles. For this reason we work with the coefficients $\mathbf{Z}/\ell^n$ and pass to E only at the end. Apart from this, the only arithmetic input is the finiteness of $J_0(k_0)$, which holds because J_0 is of finite type over a finite field.

Lemma 14.1 (A proper smooth model)**.** *A smooth geometrically connected curve U_0/k_0 has a smooth projective geometrically connected compactification $j_0 : U_0 \hookrightarrow C_0$. The reduced boundary D_0 is finite étale over k_0.*

Proof. We compactify by Nagata's theorem and then normalize. On a curve, normality is the same as regularity, and over a perfect field regularity is smoothness, so the normalization is the model we want.

By Theorem A.2*, U_0 is an open subscheme of a proper k_0-scheme, and we take the reduced closure of U_0 in it and normalize this closure in $k_0(U_0)$. Since a scheme of finite type over a field is Nagata, the normalization is finite [Stacks, 035B, 035S], and it is an isomorphism over the normal open U_0. The resulting proper normal curve is regular, because a one-dimensional noetherian normal local domain is a discrete valuation ring. Hence it is smooth over the perfect field k_0 [Stacks, 0B8Y]. Its boundary is zero-dimensional and proper over k_0, hence finite. Since k_0 is perfect, the reduced boundary is étale. After base change to k, the curve U is dense and connected, and therefore its closure C is connected. Finally, a proper scheme of dimension one over a field is projective [Stacks, 0A26]. ∎

Once we have a smooth projective model C_0, the Jacobian J_0 of C_0 exists (Appendix C*), and we can compare the cohomology of C with its Tate module. We first make the comparison with

finite coefficients, where Kummer theory applies directly. The proposition ends with a matrix description of Frobenius, which is the form used in Proposition 14.3.

Proposition 14.2 (Kummer, duality and the integral Tate module)**.** *Let C_0/k_0 be a smooth proper geometrically connected curve with Jacobian J_0, and put $A_n = \mathbf{Z}/\ell^n$. There are canonical compatible Galois-equivariant identifications*

$$H^1(C, A_n(1)) \simeq J[\ell^n](k), \qquad \varprojlim_n H^1(C, A_n) \simeq \mathrm{Hom}_{\mathbf{Z}_\ell}(T_\ell J, \mathbf{Z}_\ell).$$

Here the transition maps are multiplication by ℓ on $J[\ell^n](k)$ and reduction of coefficients on $H^1(C, A_n)$. If arithmetic Frobenius on $T_\ell J$ has matrix B, then geometric Frobenius on $\mathrm{Hom}_{\mathbf{Z}_\ell}(T_\ell J, \mathbf{Z}_\ell)$ has matrix B^t in the dual basis.

Proof. By the Kummer sequence, $H^1(C, \mu_{\ell^n})$ is the ℓ^n-torsion of the Picard group, and Poincaré duality turns this into a statement about the untwisted H^1.

The first identification is Proposition C.17*, which also gives its Galois equivariance, and it shows that the coefficient map $z \mapsto z^\ell$ from $\mu_{\ell^{n+1}}$ to μ_{ℓ^n} corresponds to $[\ell]$ on the Jacobian.

For the dual Tate module, Poincaré duality at finite level pairs the untwisted H^1 with the twisted one that we have just identified with $J[\ell^n](k)$. Derived duality gives a statement about the single group H^1 because all the groups involved are free A_n-modules. Indeed, we have $H^0(C, A_n) = A_n$ and $H^2(C, A_n(1)) = A_n$ by Proposition D.2*, while the group $H^1(C, A_n(1))$ is free of rank $2g$ by the identification above and Corollary C.15*. Since $A_n(1) \simeq A_n$ over k, the untwisted H^1 is free of rank $2g$ as well. Finally, a bounded complex with projective cohomology modules is the sum of its shifted cohomology modules in the derived category, so A_n-duality acts degree by degree. In this situation derived Poincaré duality, Theorem 10.12*, applied to the constant sheaf, induces in degree one an isomorphism

$$H^1(C, A_n) \xrightarrow{\sim} \mathrm{Hom}_{A_n}(H^1(C, A_n(1)), A_n).$$

This map is the adjoint of cup product followed by the trace map of the duality theorem, and both cup product and trace commute with reduction of coefficients and with the action of $\mathrm{Gal}(k/k_0)$. Because of the twist on the second factor, Frobenius acts trivially on the target A_n.

In the limit, the finite-level dualities combine into a duality with the Tate module. Since the transition maps $[\ell]$ are surjective, we have $J[\ell^n](k) = T_\ell J/\ell^n T_\ell J$. Moreover $T_\ell J$ is finite free by Proposition B.14*, and a homomorphism of finite free $\mathbf{Z}_\ell$-modules is determined by its reductions modulo all ℓ^n, and every compatible system of reductions lifts. Hence the inverse limit of the adjoints at finite level is the dual of $T_\ell J$. We obtain a perfect pairing

$$\varprojlim_n H^1(C, A_n) \times T_\ell J \longrightarrow \mathbf{Z}_\ell,$$

invariant under Frobenius. Since geometric Frobenius acts on $T_\ell J$ by B^{-1}, its matrix on $\varprojlim_n H^1(C, A_n)$ in the dual basis is $(B^{-1})^{-t} = B^t$. ∎

For the elliptic curve $y^2 = x^3 - x$ over $\mathbf{F}_3$ of the introduction, arithmetic Frobenius B has eigenvalues $\pm i\sqrt{3}$ on $T_\ell J$. By the proposition, geometric Frobenius acts on $\varprojlim_n H^1(C, A_n)$ with the same eigenvalues. Since $\det(B^t - 1) = 4 \neq 0$, it has no fixed vector. The next proposition proves the same for every smooth curve, including open ones.

Proposition 14.3 (No fixed class in degree one)**.** *For a smooth geometrically connected curve U_0/k_0, geometric Frobenius has no fixed vector on $H^1(U, E)$ for every finite $E/\mathbf{Q}_\ell$. More precisely,*

$$\left(\varprojlim_n H^1(U, \mathbf{Z}/\ell^n)\right)^{F=1} = 0.$$

Proof. On the compactification the vanishing holds because $J_0(k_0)$ is finite, and on the boundary it holds because Frobenius acts by a power of q, which is not 1. The residue sequence reduces the open case to the proper one, and at the end we pass from integral to rational coefficients.

The proper case. Let $U_0 = C_0$ be proper, with Jacobian J_0. Here the cohomology is the dual of the Tate module, and the question is whether 1 is an eigenvalue of Frobenius on it. By Proposition B.15*, arithmetic Frobenius has no fixed vector on $T_\ell J$; indeed, a fixed compatible tuple would consist of ℓ-power torsion points of the finite group $J_0(k_0)$. This means that $B - 1$ is invertible on $V_\ell J$. On the other side, geometric Frobenius acts on the untwisted H^1 by B^t (Proposition 14.2). Since $\det(B^t - 1) = \det(B - 1) \neq 0$, it fixes neither a nonzero rational vector nor a nonzero integral compatible tuple. The argument applies to every power of Frobenius, so the same holds over every finite extension of k_0, which we need in the open case.

The open case. A fixed class has zero residues at the boundary points, so it extends to the compactification, where the proper case shows that it is zero. To make this precise, choose $j : U \hookrightarrow C$ as in Lemma 14.1, with boundary $i : D \hookrightarrow C$. At each finite level there is then a canonical residue sequence

$$0 \longrightarrow H^1(C, A_n) \longrightarrow H^1(U, A_n) \longrightarrow H^0(D, A_n(-1)).$$

To obtain it with the correct twist and Frobenius action, we use the dualizing complexes. By transitivity, $i^!\omega_C = \omega_D = A_n$. Since $\omega_C = A_n(1)[2]$, it follows that $i^!A_{n,C} = A_{n,D}(-1)[-2]$. The localization triangle $i_*i^!A_{n,C} \to A_{n,C} \to Rj_*A_{n,U}$ now yields the displayed sequence, and all its maps are natural in the coefficients and under automorphisms of the model.

Choose $m > 0$ such that F^m fixes all geometric boundary points, and put $Q = q^m$. On $\varprojlim_n H^0(D, A_n(-1))$, which is finite free over $\mathbf{Z}_\ell$, the operator F^m acts as Q. Since $Q - 1 \neq 0$, it has no nonzero fixed element. A fixed compatible class on U therefore has zero residues at every level, so by exactness it lifts uniquely to $H^1(C, A_n)$. By uniqueness the lifts are compatible and fixed by F^m, and the proper case over $\mathbf{F}_Q$ shows that they vanish.

Rational coefficients. To pass from compatible systems of classes to $H^1(U, E)$, we identify $H^1(U, \mathbf{Q}_\ell)$ with the rationalization of the inverse limit and check that rationalizing does not create fixed vectors. By Proposition 2.55*, applied to the constant sheaves A_n, we have $R\Gamma(U, \mathbf{Z}_\ell) = R\varprojlim_n R\Gamma(U, A_n)$, and hence an exact sequence

$$0 \to \varprojlim{}^1_n H^0(U, A_n) \to H^1(U, \mathbf{Z}_\ell) \to \varprojlim_n H^1(U, A_n) \to 0.$$

The $\varprojlim^1$-term vanishes because the groups $H^0(U, A_n) = A_n$ have surjective transition maps. On $U_{\text{pro\'et}}$ the sheaf $\mathbf{Q}_\ell$ is $\mathbf{Z}_\ell[1/\ell]$ (Lemma 2.51*), a filtered colimit of copies of $\mathbf{Z}_\ell$, and cohomology on the quasi-compact and quasi-separated scheme U commutes with filtered colimits of sheaves (Corollary 2.29*, applied to sheaves placed in degree zero). So $H^1(U, \mathbf{Q}_\ell) = H^1(U, \mathbf{Z}_\ell) \otimes \mathbf{Q}_\ell$ is the rationalization of the inverse limit.

The inverse limit of the residue sequences is exact on the left, and rationalization is exact, so we obtain an exact sequence of finite-dimensional $\mathbf{Q}_\ell$-spaces $0 \to H^1(C, \mathbf{Q}_\ell) \to H^1(U, \mathbf{Q}_\ell) \to I \to 0$, where I is the image of the residue map inside the rationalized boundary term. By the proper

case, $F^m - 1$ is invertible on $H^1(C, \mathbf{Q}_\ell)$. On the boundary term, $F^m - 1$ acts by $Q - 1$, so it is injective on the F^m-stable subspace I. Now let v be a vector of $H^1(U, \mathbf{Q}_\ell)$ killed by $F^m - 1$. Its image in I is killed by $F^m - 1$, hence zero. Thus v comes from $H^1(C, \mathbf{Q}_\ell)$, where it is zero by the proper case. Since a vector fixed by F is fixed by F^m, it follows that Frobenius has no fixed rational vector, and extending scalars to E gives the assertion over E. ∎

14.1.2 Finite monodromy

With Proposition 14.3 in hand, the logarithm argument sketched at the start of Section 14.1 proves Theorem 14.5. Corollaries Corollary 14.6 and Corollary 14.8 then translate finiteness of monodromy into weights. For a lisse sheaf of higher rank, however, the determinant weight (Definition 14.7) is the only weight we can attach until Rankin's argument. Throughout, U_0 is a smooth geometrically connected curve over k_0 and $s \in W(U_0)$ is a lift of degree one, as in Convention `situation-weil`. In this setting, the first step identifies the additive characters to which the logarithm leads.

Lemma 14.4 (Invariant additive characters)**.** *Every continuous additive homomorphism $a : \pi_1(U) \to E$ satisfying $a(s^{-1}gs) = a(g)$ is zero.*

Proof. After scaling, the coordinates of a reduce modulo ℓ^n to a compatible system of Frobenius-fixed classes in $H^1(U, \mathbf{Z}/\ell^n)$, which vanish by Proposition 14.3. This needs H^1 to be identified with homomorphisms.

Recall that for a finite abelian group G the group $H^1(U, G)$ classifies G-torsors. By [Stacks, 0DV5], finite locally constant sheaves on the connected scheme U correspond to finite continuous $\pi_1(U)$-sets, and under this correspondence a G-torsor becomes a finite $\pi_1(U)$-set with a commuting simply transitive action of G, that is, a continuous homomorphism $\pi_1(U) \to G$. Hence $H^1(U, G) = \mathrm{Hom}_{\mathrm{cont}}(\pi_1(U), G)$. In this description Frobenius acts on $H^1(U, G)$ through the action of s on $\pi_1(U)$ by conjugation. The Frobenius-fixed classes are therefore the homomorphisms b with $b(s^{-1}gs) = b(g)$.

Choose a $\mathbf{Q}_\ell$-basis of E, with coordinate maps t_i. Since $\pi_1(U)$ is compact, each $t_i a(\pi_1(U))$ is bounded, and after multiplying all coordinates by one power of ℓ we may assume that they take values in $\mathbf{Z}_\ell$. Each coordinate is then a continuous homomorphism $\pi_1(U) \to \mathbf{Z}_\ell$ invariant under conjugation by s. Its reductions modulo ℓ^n are compatible classes in $H^1(U, \mathbf{Z}/\ell^n)$ fixed by Frobenius. By the proposition they vanish, and hence every coordinate of a is zero. ∎

The invariance hypothesis of the lemma is automatic for characters of the Weil group. If $\chi : W(U_0) \to E^\times$ is a character, then $\chi(s^{-1}gs) = \chi(s)^{-1}\chi(g)\chi(s) = \chi(g)$, because $E^\times$ is commutative. So the lemma applies as soon as χ has been made additive on $\pi_1(U)$.

Theorem 14.5 (Rank-one sheaves have finite geometric monodromy)**.** *Let U_0/k_0 be a smooth geometrically connected curve, and let $\chi : W(U_0) \to E^\times$ be the character of a rank-one lisse Weil sheaf. Then $\chi(\pi_1(U))$ is finite.*

Proof. We show that a fixed power of χ is trivial on $\pi_1(U)$, by applying Lemma 14.4 to its logarithm. For this we need a subgroup $1 + \mathfrak{m}_E^r$ of finite index in $\mathcal{O}_E^\times$ on which the logarithm is an injective homomorphism, and a power of χ that lands in it.

Step 1: the logarithm. The compact group $\chi(\pi_1(U))$ lies in $\mathcal{O}_E^\times$, because its image under the normalized valuation is a finite subgroup of $\mathbf{Z}$ and therefore zero. Now write $e = v_E(\ell)$ and choose an integer $r > e/(\ell - 1)$. We claim that log is injective on $1 + \mathfrak{m}_E^r$. For $x \in \mathfrak{m}_E^r$ and $n \geq 1$ we have $v_E(n) = e\, v_\ell(n)$ and, by Legendre's formula, $v_\ell(n!) \leq (n-1)/(\ell - 1)$, so that

$$v_E(x^n/n) \geq nr - e\,v_\ell(n), \qquad v_E(x^n/n!) \geq r + (n-1)(r - e/(\ell-1)).$$

Both bounds tend to infinity with n and are at least r: for the second this is clear, and for the first it follows from $n-1 \geq \ell^{v_\ell(n)} - 1 \geq (\ell-1)v_\ell(n)$ and $(\ell-1)r > e$, which give $(n-1)r \geq e\,v_\ell(n)$. The logarithm and exponential series therefore converge on $\mathfrak{m}_E^r$; the logarithm maps $1+\mathfrak{m}_E^r$ into $\mathfrak{m}_E^r$, and the exponential maps $\mathfrak{m}_E^r$ into $1+\mathfrak{m}_E^r$. By the usual identities of formal power series, $\log : 1+\mathfrak{m}_E^r \to \mathfrak{m}_E^r$ and exp are mutually inverse continuous group homomorphisms.

Step 2: a power of χ. The quotient $\mathcal{O}_E^\times/(1+\mathfrak{m}_E^r)$ is finite, being isomorphic to $(\mathcal{O}_E/\mathfrak{m}_E^r)^\times$; let $N > 0$ be an integer that kills it. As $\chi(\pi_1(U)) \subset \mathcal{O}_E^\times$, we have $\chi(g)^N \in 1+\mathfrak{m}_E^r$ for $g \in \pi_1(U)$. Hence $a(g) = \log(\chi(g)^N)$ defines a continuous homomorphism $\pi_1(U) \to E$, which is invariant under conjugation by s because χ takes values in an abelian group.

Step 3: conclusion. Lemma 14.4 gives $a = 0$, so $\chi(g)^N = 1$ for every $g \in \pi_1(U)$, the logarithm being injective. The theorem follows, since $T^N - 1$ has only finitely many roots in E. ∎

Finiteness gives the sheaf a weight, and since the lift s is a choice, part of the proof checks that the weight does not depend on it.

Corollary 14.6 (Rank-one weights)**.** *Every rank-one lisse Weil sheaf on a smooth geometrically connected curve is punctually ι-pure. For $c = \chi(s)$ its weight is $\lambda_\iota(c) = 2\log_q|\iota(c)|$.*

Proof. We separate from χ the part that depends only on the degree. Put $\varepsilon(w) = \chi(w)c^{-\deg w}$, a character of $W(U_0)$ that is trivial on s. Every element of $W(U_0)$ can be written as $w = gs^d$ with $g \in \pi_1(U)$ and $d = \deg w$, and then $\varepsilon(w) = \chi(g)\chi(s)^d c^{-d} = \chi(g)$. So ε has the same image as $\chi|_{\pi_1(U)}$, which is finite by Theorem 14.5. At a closed point x the eigenvalue is $\varepsilon(\mathrm{Frob}_x)c^{\deg(x)}$, where the first factor has finite order, hence complex absolute value one. Therefore the eigenvalue has absolute value $q_x^{\lambda_\iota(c)/2}$. This number does not depend on s, because changing s multiplies c by a value of χ on $\pi_1(U)$, which has finite order. ∎

For a Kummer sheaf on $\mathbf{G}_{m,k_0}$ the corollary gives weight 0, since $\chi(s)$ is then a root of unity. For $\mathcal{B}_b$ it gives $\lambda_\iota(b)$, the weight found in Proposition 13.7*.

Before purity is known, the eigenvalues of a lisse sheaf $\mathcal{V}$ of rank r at a closed point are not controlled individually. Their product, however, is controlled, since it is the eigenvalue of the rank-one sheaf $\det\mathcal{V}$, whose weight is given by Corollary 14.6. Dividing by r we get the average weight of the eigenvalues, which is the only candidate for a weight of $\mathcal{V}$. Indeed, if $\mathcal{V}$ is pure of weight w, then $\det\mathcal{V}$ is pure of weight rw, and the number defined below is w. For a sheaf that is not pure, on the other hand, the number is only an average. For example, on $\overline{\mathbf{Q}}_\ell \oplus \overline{\mathbf{Q}}_\ell(-1)$ the determinant is $\overline{\mathbf{Q}}_\ell(-1)$, of weight 2, and the number is 1, the weight of no eigenvalue. For this reason we apply it to irreducible constituents: Proposition 14.25 bounds the eigenvalues of a real sheaf by the largest determinant weight of its constituents, and Theorem 14.26 then shows that each irreducible constituent is pure of weight equal to its determinant weight.

Definition 14.7 (Determinant weight)**.** Let $\mathcal{V}$ be a nonzero lisse Weil sheaf of rank r on a smooth geometrically connected curve U_0, let ρ be the corresponding representation of $W(U_0)$, and let $s \in W(U_0)$ be a lift of degree one. The *determinant weight* of $\mathcal{V}$ is

$$\mathrm{dw}_\iota(\mathcal{V}) = \frac{\lambda_\iota(\det\rho(s))}{r}$$

(this does not depend on s, Corollary 14.8). For a lisse Weil sheaf $\mathcal{V}$ we write $\mathrm{DW}(\mathcal{V})$ for the set of determinant weights of its irreducible constituents.

Corollary 14.8 (Determinant weights)**.** *Let $\mathcal{V}$ be a nonzero lisse Weil sheaf of rank r on a smooth geometrically connected curve. Its determinant has finite geometric monodromy and is punctually pure of weight $r\,\mathrm{dw}_\iota(\mathcal{V})$. The number $\mathrm{dw}_\iota(\mathcal{V})$ does not depend on the choice of s or of a finite coefficient field over which $\mathcal{V}$ is defined.*

Proof. The sheaf $\det\mathcal{V}$ is a rank-one lisse Weil sheaf defined over a finite extension of E, so Theorem 14.5 and Corollary 14.6 apply to it. By the formula of Corollary 14.6, with $c = \det\rho(s)$, the weight of $\det\mathcal{V}$ is $r\,\mathrm{dw}_\iota(\mathcal{V})$, and it does not depend on s. Finally, a finite extension of scalars changes neither the character nor its absolute value under the fixed embedding. ∎

The corollary applies to every irreducible constituent of $\mathcal{V}$ and of every tensor or exterior power of $\mathcal{V}$. Indeed, a finite-dimensional representation has a composition series, and each factor is again lisse and defined over a finite extension of $\mathbf{Q}_\ell$, because after enlarging E the invariant subspaces and quotients have bases defined over E.

Remark 14.9 (Determinant weight of a sheaf with pure constituents)**.** Let $\mathcal{V}$ be a nonzero lisse Weil sheaf of rank r on a smooth geometrically connected curve. If the irreducible constituents $\mathcal{V}_1,\dots,\mathcal{V}_n$ of $\mathcal{V}$ are pure of weights $w_1,\dots,w_n$, then $\mathrm{dw}_\iota(\mathcal{V}) = \sum_a \mathrm{rank}(\mathcal{V}_a)w_a/r$. In particular a sheaf pure of weight w has determinant weight w.

14.2 Determinant weights

Before purity is known, the only weight we can attach to an irreducible lisse Weil sheaf Q on a curve is its determinant weight $\mathrm{dw}_\iota(Q)$, the weight of $\det Q$ divided by the rank (Definition 14.7 and Corollary 14.8). This section proves the two facts about determinant weights that Rankin's argument in Section 14.3 needs for the tensor and exterior powers of a real sheaf $\mathcal{G}$. First, every simple factor of $\mathcal{G}^{\otimes 2m}$ has determinant weight at most $2m$ times the largest determinant weight of the constituents of $\mathcal{G}$, and this bounds the weights of H^2_c. Second, the determinant weights of the simple factors of $\bigwedge^r\mathcal{G}$ are known exactly, and this is what lets exterior powers separate the constituents.

The difficulty is that a determinant only sees an average. For irreducible A and B of ranks a and b, we have $\det(A\otimes B) = (\det A)^{\otimes b}\otimes(\det B)^{\otimes a}$. Hence the determinant weight of $A\otimes B$ as a whole is $\mathrm{dw}_\iota(A)+\mathrm{dw}_\iota(B)$. But $A\otimes B$ is usually reducible, and which subquotients occur depends on the monodromy. The rank-weighted average of the determinant weights of its simple factors is the number above, but nothing so far prevents one factor from lying above it and another below. If the factors were pure, the determinant weight of each would be its weight, but purity is what we are trying to prove.

The model to keep in mind is a central character. If a group acts on V and some central element z acts on V by the scalar c, then z acts on every subquotient of $V^{\otimes a}\otimes(V^\vee)^{\otimes b}$ by c^{a-b}, without any knowledge of how these representations decompose. We therefore look for a central element of positive degree in the monodromy group; its scalars will record weights.

Let $H\subseteq G$ be the Zariski closures in $\mathrm{GL}(V)$ of the images of $\pi_1(U)$ and of $W(U_0)$, the geometric and arithmetic monodromy groups. If H were trivial, the image $\rho(s)$ of a degree-one lift would itself be central. In general we use the structure of the identity component,

$$G^\circ = Z(G^\circ)^\circ H^\circ$$

for V semisimple (Lemma 14.15). The reason is that the Lie algebra of G° is reductive, hence the sum of its center and its derived algebra, and since G/H is abelian, the derived algebra lies

in the Lie algebra of H. Accordingly, in Proposition 14.17 we write a power of $\rho(s)$ as a central torus element times an element of H, and then we take the product of its conjugates over G/G°. The result is an element $z = h\rho(s)^N$, central in G, with $h \in H$ and $N > 0$.

On a simple factor Q of any tensor construction, z acts by a scalar c_Q, and taking determinants gives $c_Q^{\dim Q} = \det(h \mid Q)\det(s \mid Q)^N$. Since determinants have finite geometric monodromy, $\det(h \mid Q)$ is a root of unity, and from this Lemma 14.18 concludes that $\mathrm{dw}_\iota(Q) = \lambda_\iota(c_Q)/N$, where $\lambda_\iota(a) = 2\log_q|\iota(a)|$ is the additive function of Proposition 13.7*. Since scalars multiply under tensor products, we obtain Proposition 14.19.

The facts on algebraic groups that we use, namely the relation between closed subgroups, their identity components and their Lie algebras, and the structure of a Lie algebra with a faithful semisimple representation, are proved in Appendix F*.

14.2.1 Representations and filtrations

We treat lisse sheaves as representations of the Weil group. From the geometry of the curve we use only one fact, proved in Section 14.1: the geometric fundamental group acts through a finite group on every determinant. Everything else is linear algebra and the structure of algebraic groups.

Throughout, U_0 is a smooth geometrically connected curve over k_0 and $s \in W(U_0)$ is a lift of degree one, as in Convention `situation-weil`, so that $W(U_0) = \pi_1(U) \rtimes \langle s \rangle$.

Definition 14.10 (Weil representation)**.** A *Weil representation* is a finite-dimensional $\overline{\mathbf{Q}}_\ell$-representation of $W(U_0)$ whose restriction to $\pi_1(U)$ is continuous.

By Convention `situation-weil` the Weil representations are the lisse Weil $\overline{\mathbf{Q}}_\ell$-sheaves on U_0, and so, by Corollary 14.8, $\pi_1(U)$ acts through a finite group on the determinant of every Weil representation. A Weil representation V is semisimple if it is a direct sum of simple representations, and the semisimplification V^{ss} of V is the direct sum of its composition factors.

Two elementary facts are needed. First, semisimplicity is preserved by restriction to a normal subgroup, which lets us apply Lie theory to the identity component of the monodromy group below. Second, the composition factors of tensor and exterior powers are computed from those of the given representations, so it suffices to prove the formulas for semisimple representations.

Lemma 14.11 (Restriction to a normal subgroup)**.** *Let N be a normal subgroup of a group Γ. The restriction to N of a finite-dimensional semisimple $\overline{\mathbf{Q}}_\ell$-representation of Γ is semisimple.*

Proof. It suffices to consider a simple Γ-module V. A nonzero N-submodule A of least dimension is simple, and since N is normal, each translate gA is again a simple N-submodule. The sum of all these translates is a nonzero Γ-submodule, hence equals V. It remains to see that a finite sum of simple submodules is semisimple. Indeed, the intersection of a simple submodule with the sum of those already chosen is either zero or the whole submodule, so we add it in the first case and omit it in the second, and after finitely many steps we obtain a direct-sum decomposition. ∎

The second fact concerns representations that are not semisimple. A nonsplit extension of $\overline{\mathbf{Q}}_\ell(-1)$ by $\overline{\mathbf{Q}}_\ell$, such as the twist $\mathcal{E}(-1)$ of the Kummer extension of Example 13.4*, does not decompose. Yet the composition factors of its tensor and exterior powers are determined by those of the sheaf, as the next lemma shows by filtering. This is what lets us pass to semisimplifications in Proposition 14.19.

Lemma 14.12 (Tensor and exterior powers of filtered representations)**.** *Suppose V and V' have composition factors $A_1, \dots, A_t$ and $B_1, \dots, B_u$. There is a $W(U_0)$-stable filtration of $V \otimes V'$ with graded pieces $A_a \otimes B_b$, one for each pair (a, b). There is a $W(U_0)$-stable filtration of $\bigwedge^r V$ with graded pieces*

$$\bigotimes_{a=1}^{t} \bigwedge^{n_a} A_a, \qquad 0 \le n_a \le \dim A_a, \qquad \sum_a n_a = r.$$

The simple factors of $V \otimes V'$ and $\bigwedge^r V$, counted with multiplicity, are those of the respective graded pieces. Every graded piece of $\bigwedge^r V$ displayed above is nonzero.

Proof. We construct the two filtrations and then compare simple factors by the Jordan–Hölder argument. The tensor product is filtered by the filtration on V, and each quotient $A_a \otimes V'$ by the filtration on V'; since tensoring over a field is exact, the quotients are as asserted.

For the exterior power we cannot simply filter factor by factor, because a wedge product may use several vectors from the same composition factor. Instead we sort wedges of basis vectors by how many vectors they take from each factor, after lifting bases of the A_a to a basis of V adapted to its filtration. Thus a wedge of basis vectors determines a tuple (n_a), by counting how many vectors it uses from each lifted basis. Since the basis is adapted, the group $W(U_0)$ carries a lifted basis vector of A_a into the span of the lifted bases of $A_1, \dots, A_a$. Hence the action is block upper triangular: each term of the image of a basis wedge either has the same tuple or has a tuple obtained from it by moving some of its vectors to smaller indices a. Choose a total order on the tuples compatible with these moves. Then the spans of the initial segments of basis wedges form a stable filtration, and on the quotient indexed by (n_a) the diagonal blocks act as on $\bigotimes_a \bigwedge^{n_a} A_a$, a space of dimension $\prod_a \binom{\dim A_a}{n_a} > 0$. For example, for V an extension of A_2 by A_1, both of dimension one, and $r = 1$, the tuples are $(1, 0)$ and $(0, 1)$, and the filtration is the given one. For $r = 2$ there is the single tuple $(1, 1)$, and $\bigwedge^2 V = A_1 \otimes A_2$.

The graded pieces of the filtrations just built need not be simple, so to compare simple factors we refine. By the Schreier refinement theorem, any two finite filtrations have refinements with isomorphic graded pieces, matched by the Zassenhaus isomorphism $(V_{i-1} + V_i \cap W_j)/(V_{i-1} + V_i \cap W_{j-1}) \simeq (W_{j-1} + W_j \cap V_i)/(W_{j-1} + W_j \cap V_{i-1})$; for two composition series this is the Jordan–Hölder theorem. Applying the same refinement to a composition series and to the tensor or exterior filtration, we obtain the statement about simple factors. In particular, these multisets do not change when V and V' are replaced by their semisimplifications. ∎

14.2.2 Central elements

In the arithmetic monodromy group we construct a central element of positive degree (Proposition 14.17). It acts by a scalar on each simple factor, and this scalar determines the determinant weight of the factor. As explained above, we construct the element for semisimple representations, and then use stable filtrations to obtain the formulas for arbitrary representations.

The degree must be positive. Indeed, an element of H that is central in G also acts by a scalar on each simple factor Q. But, as for every element of H, its determinant on Q is a root of unity, so under ι the scalar has absolute value one, and it records no weight. By contrast, an element $z = h\rho(s)^N$ with $N > 0$ contains N factors of Frobenius, and its scalar c_Q on Q satisfies $\lambda_\iota(c_Q) = N \deg_\iota(Q)$ (Lemma 14.18).

We produce the central element by Lie theory, which applies to algebraic groups and not to the abstract images of $\pi_1(U)$ and $W(U_0)$. For $\mathcal{B}_b$ with b not a root of unity, the image of $W(U_0)$ in GL_1 is the infinite cyclic group generated by b, which is not an algebraic subgroup; its Zariski

closure is all of GL_1, since the proper closed subgroups of $\mathbf{G}_m$ are finite. Passing to the closure loses nothing that we need. Indeed, stabilizers are closed, so a subspace of V is stable under a group of matrices if and only if it is stable under its Zariski closure. For this reason Lemma 14.15 and Proposition 14.17 are statements about these closures.

Definition 14.13 (Monodromy groups)**.** Let $\rho : W(U_0) \to \mathrm{GL}(V)$ be a Weil representation. Its *geometric* and *arithmetic monodromy groups* are the Zariski closures H of $\rho(\pi_1(U))$ and G of $\rho(W(U_0))$ in $\mathrm{GL}(V)$.

We keep the letters H and G for these two groups in the rest of this section.

Lemma 14.14 (Normality of geometric monodromy)**.** *The groups H and G are reduced closed algebraic subgroups of $\mathrm{GL}(V)$, and $H \triangleleft G$. Moreover every commutator of two elements of G lies in H.*

Proof. Each assertion holds for the images of $\pi_1(U)$ and $W(U_0)$, and we pass to Zariski closures by continuity. First, multiplication and inversion are continuous for the Zariski topology, so the closure of a subgroup is a subgroup. Next, conjugation by $\rho(W(U_0))$ preserves H because $\pi_1(U)$ is normal in $W(U_0)$, and since the normalizer of a closed subgroup is closed, conjugation by G preserves H as well. Finally, since $W(U_0)/\pi_1(U) \simeq \mathbf{Z}$ is abelian, commutators of elements of $\rho(W(U_0))$ lie in $\rho(\pi_1(U))$. As the commutator map $G \times G \to \mathrm{GL}(V)$ is continuous, the same holds on the closures. ∎

For example, for $V = \mathcal{B}_b$ of Proposition 13.7* the group H is trivial. Hence $\rho(s) = b$ is central in G, and the determinant weight of $\mathcal{B}_b$ is $\lambda_\iota(b)$.

Lemma 14.15 (The identity component)**.** *If V is a semisimple $W(U_0)$-representation, then*

$$G^\circ = Z(G^\circ)^\circ H^\circ.$$

Proof. A connected algebraic group is determined by its Lie algebra among its closed connected subgroups, so it suffices to show that the Lie algebras of $Z(G^\circ)^\circ$ and H° together span that of G°. We show that this Lie algebra is reductive, hence the sum of its center and its derived algebra, and that the derived algebra lies in the Lie algebra of H because commutators lie in H.

Reductivity. Write $\mathfrak{g} = \mathrm{Lie}(G^\circ)$. Then

$$\mathfrak{g} = Z(\mathfrak{g}) + [\mathfrak{g}, \mathfrak{g}],$$

by Proposition F.8*, once we know that V is a faithful semisimple $\mathfrak{g}$-module. The $\rho(W(U_0))$-stable subspaces are precisely the G-stable subspaces, because the stabilizer of a subspace is closed. Hence V is semisimple as a $G(\overline{\mathbf{Q}}_\ell)$-module, and by Lemma 14.11 it is also semisimple as a $G^\circ(\overline{\mathbf{Q}}_\ell)$-module. Then Lemma F.3* makes it a faithful semisimple $\mathfrak{g}$-module.

The center and the derived algebra. Put $T = Z(G^\circ)^\circ$ and $\mathfrak{h} = \mathrm{Lie}(H^\circ)$. The comparison between groups and Lie algebras in Lemma F.3* gives $\mathrm{Lie}(T) = Z(\mathfrak{g})$. Moreover $[\mathfrak{g}, \mathfrak{g}] \subseteq \mathfrak{h}$, because the commutator morphism $G \times G \to G$ factors through the closed subgroup H: it does so on $\overline{\mathbf{Q}}_\ell$-points by Lemma 14.14, and $G \times G$ is reduced. To deduce the inclusion, let $X, Y \in \mathfrak{g}$ and put $R = \overline{\mathbf{Q}}_\ell[\varepsilon, \eta]/(\varepsilon^2, \eta^2)$. The commutator of $1 + \varepsilon X$ and $1 + \eta Y$ in $G(R)$ is $1 + \varepsilon\eta[X, Y]$, and it lies in $H(R)$. Here the subring $\overline{\mathbf{Q}}_\ell[\varepsilon\eta] \subset R$ is a ring of dual numbers, and the equations of H vanish on $1 + \varepsilon\eta[X, Y]$ already over this subring. Hence $[X, Y] \in \mathrm{Lie}(H) = \mathfrak{h}$, and

$$\mathfrak{g} = \mathrm{Lie}(T) + \mathfrak{h}.$$

Back to groups. The multiplication $T \times H^\circ \to G^\circ$ is a homomorphism, because T is central. By Lemma F.2* its image is a closed connected subgroup, whose Lie algebra contains both summands on the right, hence equals that of G°. By the same lemma the image is therefore G°. ∎

Rankin's argument needs the determinant weights of the simple factors of $\mathcal{G}^{\otimes 2m}$, of the exterior powers of $\mathcal{G}$ and of the dual, so we give a name to the representations built in this way; the monodromy group acts on all of them.

Definition 14.16 (Tensor construction)**.** A *tensor construction* on a Weil representation V is a representation obtained from V and $V^\vee$ by finitely many tensor products, direct sums, exterior powers and subquotients by $W(U_0)$-stable subspaces.

Since the stabilizer of a subspace is closed, a $W(U_0)$-stable subspace is G-stable, and so G acts on every tensor construction through its action on V.

Proposition 14.17 (A central element of the arithmetic monodromy group)**.** *Let $V_1, \dots, V_n$ be semisimple Weil representations and form the monodromy groups $H \subseteq G$ of $V = \bigoplus_i V_i$, and let $s \in W(U_0)$ have degree one. There are an integer $N > 0$ and $h \in H(\overline{\mathbf{Q}}_\ell)$ such that*

$$z = h\,\rho(s)^N \in Z(G)(\overline{\mathbf{Q}}_\ell).$$

The action of z on each tensor construction on V preserves every $W(U_0)$-stable subquotient and is scalar on each simple subquotient.

Proof. We write a power of $\rho(s)$ as a central torus element times an element of H, by Lemma 14.15, and average the torus part over the components of G so that it becomes central in all of G. The factors from H that accumulate in this process are collected in h; their determinants are roots of unity, so by Lemma 14.18 they do no harm. Note that the direct sum V of the V_i is again semisimple, so Lemma 14.15 applies to it.

Since the component group of G is finite (Lemma F.2*), we may choose $m > 0$ with $\rho(s)^m \in G^\circ$. By Lemma 14.15, we can write $\rho(s)^m = ct$ with $c \in Z(G^\circ)^\circ$ and $t \in H^\circ$. Choose representatives $g_1, \dots, g_b$ of the components of G and set

$$z = \prod_{j=1}^{b} g_j c g_j^{-1}, \qquad N = mb.$$

The element c is central in G° but perhaps not in G, since the other components may conjugate it nontrivially. The product z of its conjugates, one for each component, is however central in G. Indeed, the factors lie in the abelian group $Z(G^\circ)^\circ$, and since replacing g_j by $g_j u$ with $u \in G^\circ$ does not change the corresponding factor, conjugation by an element of G permutes the factors. Next, every commutator lies in H, so each factor has the same image as c in G/H. Since $c \in H\rho(s)^m$, we get $z \in H\rho(s)^{mb}$, which gives h and $N = mb > 0$.

Being central in G, the element z commutes with the action of $W(U_0)$ on tensor constructions. Moreover, as noted after the definition of tensor constructions, every $W(U_0)$-stable subspace of a tensor construction is G-stable. Hence z acts on the $W(U_0)$-stable subquotients and commutes with the action of $W(U_0)$ on them. On a simple subquotient, therefore, an eigenspace of z is nonzero and $W(U_0)$-stable, hence is the whole space; in other words, z acts by a scalar. ∎

14.2.3 Tensor operations

The central element acts on each simple factor of a tensor construction by a scalar, and from these scalars we read off determinant weights. Since Weil representations are the lisse Weil $\overline{\mathbf{Q}}_\ell$-sheaves, Definition 14.7 applies to them.

Lemma 14.18 (Determinant weight from a central scalar)**.** *In the setting of Proposition 14.17, let Q be a nonzero subquotient of a tensor construction on V on which the element z acts by a scalar c_Q. Then*

$$\mathrm{dw}_\iota(Q) = \frac{\lambda_\iota(c_Q)}{N}.$$

Proof. The determinant of Q is an algebraic character of H, and its image on the dense subgroup $\rho(\pi_1(U))$ is finite, and hence its image on all of H lies in the same finite group of roots of unity. Taking determinants in $z = h\rho(s)^N$, we find

$$c_Q^{\dim Q} = \det(h \mid Q)\det(s \mid Q)^N.$$

Now apply λ_ι, which vanishes on the first factor, and divide by $N \dim Q$. ∎

The lemma reduces determinant weights of tensor constructions to a computation with the scalars on the composition factors of V. By Proposition 14.17, the element z acts by a scalar on every simple subquotient of a tensor construction on V, so the formula of the lemma applies to each of them. Since the scalars by which a single element acts multiply under tensor products and exterior powers, this gives the following formulas.

Proposition 14.19 (Determinant weights of tensor operations)**.** *Let V and V' be nonzero Weil representations, let $A_1, \dots, A_t$ be the composition factors of V, and put $\gamma_a = \mathrm{dw}_\iota(A_a)$.*

1. $\mathrm{DW}(V \otimes V') = \mathrm{DW}(V) + \mathrm{DW}(V')$.
2. $\mathrm{DW}(\bigwedge^r V) = \{\sum_a n_a\gamma_a : 0 \le n_a \le \dim A_a,\ \sum_a n_a = r\}$ *for* $0 \le r \le \dim V$.
3. $\mathrm{DW}(V^\vee) = -\mathrm{DW}(V)$.

Proof. We use one central element for V and V' simultaneously, so that its scalars on the factors of $V \otimes V'$ are products of its scalars on the factors of V and of V'. Since Proposition 14.17 requires semisimple representations, we apply it to $V^{\mathrm{ss}} \oplus V'^{\mathrm{ss}}$. This changes nothing, because by Lemma 14.12 the simple factors of $V \otimes V'$ and $\bigwedge^r V$ depend only on the composition factors of V and V'. Now let c_a and c'_b be the scalars by which its central element of degree N acts on the composition factors A_a of V and A'_b of V'. By Lemma 14.18, $\gamma_a = \lambda_\iota(c_a)/N$ and $\mathrm{dw}_\iota(A'_b) = \lambda_\iota(c'_b)/N$.

(1) The central element acts by $c_a c'_b$ on $A_a \otimes A'_b$, and hence on every simple factor of it. By the same lemma, each such factor has determinant weight $\gamma_a + \mathrm{dw}_\iota(A'_b)$. Moreover, each $A_a \otimes A'_b$ is nonzero, and Lemma 14.12 identifies the simple factors of these tensor products with those of $V \otimes V'$.

(2) The central element acts on $\bigotimes_a \bigwedge^{n_a} A_a$ by $\prod_a c_a^{n_a}$. So each simple factor of this nonzero representation has determinant weight $\sum_a n_a\gamma_a$, and Lemma 14.12 identifies these factors with those of $\bigwedge^r V$.

(3) We have $\det(s \mid A^\vee) = \det(s \mid A)^{-1}$, and duality carries a composition series of V to one of $V^\vee$ whose factors are the duals $A_a^\vee$. ∎

In particular every simple factor of $V^{\otimes m}$ has determinant weight at most $m \max \mathrm{DW}(V)$. In Section 14.4 we use this bound, and part (2), for the constituents of a real sheaf before we know that they are pure.

14.3 Rankin's bound

We bound the local eigenvalues of a real sheaf by the eigenvalues of Frobenius on H^2_c of its even tensor powers (Proposition 14.25). Only H^2_c is needed, and Lemma 14.20 computes it by Poincaré duality as the coinvariants of the geometric monodromy.

Rankin's argument uses one fact from each side of the trace formula. On the cohomological side, an affine curve has $H^0_c = 0$, so only H^2_c can produce poles of the L-function, and we can compute this group without knowing anything about H^1_c: its eigenvalues are q times eigenvalues of Frobenius on a quotient of the generic fibre. On the side of the Euler product, if the local factors have nonnegative coefficients, then each of them is holomorphic wherever the product is.

For example, for the constant sheaf on $\mathbf{G}_m$ we have

$$L(\mathbf{G}_m, \overline{\mathbf{Q}}_\ell, t) = \prod_{x \in |\mathbf{G}_m|} (1 - t^{\deg(x)})^{-1} = \frac{1-t}{1-qt},$$

where $1 - t$ comes from H^1_c and the pole at $t = q^{-1}$ from $H^2_c = \overline{\mathbf{Q}}_\ell(-1)$. Each local factor has its poles on the circle $|t| = 1$, outside the disc $|t| < q^{-1}$, as Lemma 14.24 predicts. In general we apply the two facts to the even tensor powers of a real sheaf, whose H^2_c is bounded by the determinant weights of Section 14.2, and this gives Proposition 14.25.

Lemma 14.20 (Degree zero and top cohomology)**.** *Let $U_0/\mathbf{F}_q$ be a smooth geometrically connected curve, let $\mathcal{F}$ be a lisse Weil sheaf on U_0, and let V be its geometric generic fibre, a representation of the geometric fundamental group $\pi_1(U)$. There are canonical Frobenius-equivariant identifications*

$$H^0(U, \mathcal{F}) = V^{\pi_1(U)}, \qquad H^2_c(U, \mathcal{F}) = V_{\pi_1(U)}(-1).$$

There is no compactly supported cohomology outside degrees $0, 1, 2$. If U_0 is affine, $H^0_c(U, \mathcal{F}) = 0$. If $\mathcal{F}$ is punctually of weights $\leq a$, the weights of H^0 are at most a and those of H^2_c at most $a + 2$. If $\mathcal{F}$ is pure of weight a, the two groups, when nonzero, are pure of weights a and $a + 2$, respectively.

Proof. We deduce both formulas from the description of a lisse sheaf by its representation, using Poincaré duality to reduce H^2_c to H^0, and then we read off the weights from the stalks.

A global section of a local system is determined by its value at $\bar{\eta}$, and a vector of the fibre extends to a global section exactly when it is fixed by the monodromy. This is the first formula. For the second, by degreewise Poincaré duality (Corollary 10.13*) we have

$$H^2_c(U, \mathcal{F})^\vee = H^0(U, \mathcal{F}^\vee(1)) = (V^\vee)^{\pi_1(U)}(1).$$

Since the dual of $(V^\vee)^{\pi_1(U)}$ is $V_{\pi_1(U)}$, this gives the second formula, with its twist. Moreover, the identification is induced by the canonical duality pairing and is therefore Frobenius equivariant.

Duality also gives the vanishing outside degrees $0, 1, 2$. Indeed, for $i < 0$ there is nothing to prove, and for $i > 2$ the space $H^i_c(U, \mathcal{F})$ is dual to $H^{2-i}(U, \mathcal{F}^\vee(1))$, which is zero because $2 - i < 0$. For

torsion coefficients the vanishing comes instead from the bound on cohomological dimension of Appendix D*.

Suppose that U_0 is affine, and choose a smooth proper completion of U. Its boundary is nonempty, because a connected proper curve has only constant global functions and hence is not affine. A compactly supported section extends by zero to the completion, so it vanishes on a neighbourhood of the boundary. Since a section of a lisse sheaf on a connected curve is determined by its generic value, the section is zero, and $H^0_c(U, \mathcal{F}) = 0$.

For the weights we compare the global Frobenius on these two spaces with the local Frobenius at a closed point. Since $\pi_1(U)$ acts trivially on $V^{\pi_1(U)}$ and on $V_{\pi_1(U)}$, the action of $W(U_0)$ on them factors through the degree quotient. If β is an eigenvalue of the operator of degree one, then $\beta^{\deg(x)}$ is an eigenvalue of Frob_x on $\mathcal{F}_{\bar{x}}$ for every closed point x. Indeed, $V^{\pi_1(U)}$ is a subspace and $V_{\pi_1(U)}$ a quotient of $V \simeq \mathcal{F}_{\bar{x}}$, both stable under $\mathrm{Frob}_x \in \pi_1(U) s^{\deg(x)}$, which acts on them as $s^{\deg(x)}$. If $\mathcal{F}$ is punctually of weights $\leq a$, it follows that $|\iota(\beta)| \leq q^{a/2}$, and if $\mathcal{F}$ is pure of weight a, equality holds. Finally, the twist (-1) multiplies each global eigenvalue by q, which gives the stated weights for H^2_c. ∎

For the constant sheaf on $\mathbf{G}_m$ the lemma gives $H^2_c = \overline{\mathbf{Q}}_\ell(-1)$, of weight 2, as in the computation at the start of the section. Now take a Kummer sheaf of a nontrivial character on $\mathbf{G}_m$. Then $H^0_c = 0$ because $\mathbf{G}_m$ is affine, and $H^2_c = 0$ because the coinvariants vanish.

The positivity we want is $a^{2m} \geq 0$ for the traces a of all powers Frob_x^n on $\mathcal{G}$, and for this all these traces must be real. It is not enough to ask that the trace of Frob_x itself be real. For example, if the eigenvalues at x are i and $1 - i$, the trace is 1, but the trace of the square is $-1 + (1 - i)^2 = -1 - 2i$. By the Newton identities, on the other hand, all the power sums are real exactly when the characteristic polynomial is real, and this is the condition below. Its basic examples are the constant sheaf, its Tate twists and Kummer sheaves of quadratic characters. More generally, in Chapter 15*, Proposition 15.9* makes every pure sheaf a summand of a real one after a twist. The condition enters the proof of Theorem 14.26 through Lemma 14.22.

Definition 14.21 (Real sheaves)**.** A Weil sheaf $\mathcal{G}$ is ι-real if $\iota \det(1 - T \,\mathrm{Frob}_x \mid \mathcal{G}_{\bar{x}})$ has real coefficients for every closed point; equivalently, the multiset of its complex eigenvalues at each point is stable under complex conjugation. In particular, the traces of all positive powers of local Frobenius are real.

The next lemma is the ℓ-adic form of Rankin's trick. Since the even powers of a real trace are nonnegative, the Euler factors of $\mathcal{G}^{\otimes 2m}$ have nonnegative coefficients.

Lemma 14.22 (Positivity of Euler factors)**.** *Let $\mathcal{G}$ be an ι-real Weil sheaf, let $m \geq 1$, and let x be a closed point. Then the power series $\iota \det(1 - \mathrm{Frob}_x \, t^{\deg(x)} \mid \mathcal{G}_{\bar{x}}^{\otimes 2m})^{-1}$ has nonnegative real coefficients and constant coefficient one.*

Proof. We write an inverse characteristic polynomial as the exponential of a series in the traces of powers, whose coefficients are then visibly nonnegative. For a square matrix A, triangularizing A and expanding the logarithm of each factor we get the formal identity

$$\det(1 - tA)^{-1} = \exp\left(\sum_{n \geq 1} \mathrm{Tr}(A^n) \frac{t^n}{n}\right).$$

On an even tensor power, the trace of the nth power of Frobenius is $\mathrm{Tr}(\mathrm{Frob}_x^n \mid \mathcal{G})^{2m}$. Since $\mathcal{G}$ is real, this is nonnegative after applying ι. Every coefficient of the exponential is then a finite sum of products of nonnegative numbers. This remains true after replacing t by $t^{\deg(x)}$. ∎

Example 14.23 (Why even tensor powers)**.** The sheaf $\mathcal{B}_{-1}$ is real, since Frob_x acts by $(-1)^{\deg(x)}$, but its Euler factor at a point of degree one is $(1+t)^{-1} = 1-t+t^2-\cdots$, with negative coefficients. Its square $\mathcal{B}_{-1} \otimes \mathcal{B}_{-1}$ is the constant sheaf, whose Euler factors $(1-t^{\deg(x)})^{-1}$ have nonnegative coefficients, as the lemma predicts for $m = 1$.

Without positivity a factor can converge on a smaller disc than the product, as in the cancellation at the start of this chapter. With positivity, however, the coefficients of each factor are bounded by those of the product.

Lemma 14.24 (Positivity and convergence)**.** *Let*

$$L(t) = \prod_j f_j(t), \qquad f_j \in 1 + t\mathbf{R}_{\geq 0}[[t]],$$

be a coefficientwise locally finite product. If L is the Taylor series of a function holomorphic on $|t| < R$, then every f_j has radius of convergence at least R, and if $f_j = P_j^{-1}$ with $P_j(0) = 1$, then P_j has no zero in that disc.

Proof. We compare coefficients and then apply Cauchy's estimate. To compare, write $f_j = \sum_n a_{j,n} t^n$ and $L = \sum_n A_n t^n$. Since the product is coefficientwise locally finite, A_n is already the coefficient of t^n in a finite subproduct containing f_j, which is f_j times a series g with constant coefficient one and nonnegative coefficients. The coefficient of t^n in $f_j g$ is $a_{j,n}$ plus a sum of products of nonnegative numbers, so that $0 \leq a_{j,n} \leq A_n$; positivity enters only here. Next, for every $0 < r < R$, Cauchy's estimate gives $|A_n| \leq C_r r^{-n}$. Hence the radius of convergence of f_j is at least r for every such r, and so at least R. A zero of P_j in the disc would be a pole of $f_j = 1/P_j$, which is impossible. ∎

In the proposition below, the twist (-1) on H_c^2 costs a factor q, and this loss disappears when we take $2m$th roots and let m grow.

Proposition 14.25 (A bound by the largest determinant weight)**.** *Let $U_0/\mathbf{F}_q$ be a smooth geometrically connected affine curve and let $\mathcal{G} \neq 0$ be a lisse ι-real Weil sheaf. If β is the largest determinant weight of its irreducible Weil constituents, then*

$$|\iota(\alpha)|^2 \leq q_x^{\beta} \quad (x \in |U_0|,\ \alpha \text{ an eigenvalue on } \mathcal{G}_{\bar{x}}).$$

Proof. We bound the poles of $L(U_0, \mathcal{G}^{\otimes 2m}, t)$ by determinant weights through H_c^2, and positivity then passes the bound to each local factor.

We may assume that $\mathcal{G}$ is semisimple. Indeed, by Lemma 13.5*, replacing $\mathcal{G}$ by its semisimplification changes neither its local characteristic polynomials nor the determinant weights of its constituents.

The top cohomology. Fix $m \geq 1$ and put $\mathcal{T} = \mathcal{G}^{\otimes 2m}$. We show that every eigenvalue μ of F on $H_c^2(U, \mathcal{T})$ satisfies $|\iota(\mu)| \leq q^{m\beta+1}$; these eigenvalues determine the possible poles of the L-function. First, every constituent of $\mathcal{T}$ has determinant weight at most $2m\beta$, by Proposition proposition-determinant-weights-tensor-operations. Next, Lemma 14.20 identifies $H_c^2(U, \mathcal{T})$ with $T_{\pi_1(U)}(-1)$, where T is the generic fibre of $\mathcal{T}$. The geometric fundamental group acts trivially on $T_{\pi_1(U)}$, so its simple Weil factors are one-dimensional, as one sees by triangularizing the operator induced by a degree-one lift. Let b be the eigenvalue on such a factor. Since $T_{\pi_1(U)}$ is a Weil quotient of T, the factor is a constituent of $\mathcal{T}$, and its determinant weight is $\lambda_\iota(b)$. From

$\lambda_\iota(b) = 2\log_q|\iota(b)| \le 2m\beta$ we get $|\iota(b)| \le q^{m\beta}$. The twist (-1) multiplies eigenvalues by q, and the claim follows.

The local factors. The bound on H_c^2 makes the L-function holomorphic on a disc, and positivity passes this to each local factor. As $H_c^0 = 0$ by Lemma 14.20, Theorem 12.30* gives

$$L(U_0, \mathcal{T}, t) = \prod_{x \in |U_0|} \det(1 - \mathrm{Frob}_x\, t^{\deg(x)} \mid \mathcal{T}_{\bar{x}})^{-1} = \frac{\det(1 - Ft \mid H_c^1(U, \mathcal{T}))}{\det(1 - Ft \mid H_c^2(U, \mathcal{T}))}.$$

This is an identity of formal power series, and the product is coefficientwise locally finite because there are only finitely many closed points of bounded degree. Now put $R_m = q^{-m\beta-1}$. By the bound on H_c^2, the denominator has no zero on the disc $|t| < R_m$ after applying ι, so the rational function, and with it its Taylor series, is holomorphic on this disc.

Each local factor $\det(1 - \mathrm{Frob}_x\, t^{\deg(x)} \mid \mathcal{T}_{\bar{x}})$ has no zero on the same disc. Indeed, after applying ι the local factors of $L(U_0, \mathcal{T}, t)$ are power series with nonnegative coefficients and constant coefficient one by Lemma 14.22, so that Lemma 14.24 applies.

Now let α be a local eigenvalue of $\mathcal{G}$. Then α^{2m} is an eigenvalue on $\mathcal{T}$ by Lemma 13.5*, and $1 - \iota(\alpha)^{2m} t^{\deg(x)}$ divides the local factor. Its roots have absolute value $|\iota(\alpha)|^{-2m/\deg(x)}$, so by the previous paragraph this number must be at least R_m, that is,

$$|\iota(\alpha)|^2 \le q_x^{\beta + 1/m}.$$

Letting m tend to infinity proves the bound. ∎

14.4 Exterior powers and purity

Proposition 14.25 only bounds the local eigenvalues by the largest determinant weight. To bound the eigenvalues of each constituent by its own determinant weight, we apply the proposition to exterior powers, which separate the determinant weights of the simple factors.

Tensor powers cannot do this. For example, suppose $\mathcal{G} = \mathcal{G}_1 \oplus \mathcal{G}_2$ with determinant weights $\gamma_1 > \gamma_2$. In $\mathcal{G}^{\otimes m}$ the largest determinant weight $m\gamma_1$ belongs to the factors of $\mathcal{G}_1^{\otimes m}$, while every factor involving $\mathcal{G}_2$ lies strictly below it. So the bound of Proposition 14.25 says nothing sharp about $\mathcal{G}_2$.

An exterior power, by contrast, uses each basis vector of $\mathcal{G}_1$ at most once. Thus if r_1 is the rank of $\mathcal{G}_1$, every graded piece of $\bigwedge^{r_1+1} \mathcal{G}$ involves $\mathcal{G}_2$, and the summand $\det \mathcal{G}_1 \otimes \mathcal{G}_2$ has the largest determinant weight $r_1\gamma_1 + \gamma_2$. Since Corollary 14.8 gives the eigenvalues of $\det \mathcal{G}_1$, a bound for this summand is a bound for $\mathcal{G}_2$. We argue on an affine curve, where the proposition applies, and then pass to an arbitrary curve by restricting to affine opens, which preserves irreducibility.

Theorem 14.26 (Purity of real sheaves on curves)**.** *On a smooth geometrically connected curve over $\mathbf{F}_q$, every irreducible constituent of an ι-real lisse Weil sheaf is punctually ι-pure of weight equal to its determinant weight. In particular, every ι-real lisse Weil sheaf on the curve is mixed.*

Proof. We group the constituents by determinant weight and place each group, tensored with the determinants of the groups of larger weight, in an exterior power where it carries the largest determinant weight. There Proposition 14.25 bounds the eigenvalues of the group by its own determinant weight, and the same argument applied to the dual gives the reverse bound.

Let $\mathcal{G}$ be an ι-real lisse Weil sheaf on the curve. We first prove the assertion when the curve is affine. There, replacing $\mathcal{G}$ by its semisimplification changes neither its local characteristic

polynomials nor its irreducible constituents, and it keeps the sheaf real, so we may assume that $\mathcal{G}$ is semisimple and write

$$\mathcal{G} = \bigoplus_{i=1}^{t} \mathcal{G}_i, \qquad \gamma_1 > \cdots > \gamma_t, \qquad r_i = \operatorname{rank} \mathcal{G}_i,$$

where $\mathcal{G}_i$ is the sum of the irreducible constituents with determinant weight γ_i.

Fix i, and let us find the exterior power in which $\mathcal{G}_i$, together with the determinants of $\mathcal{G}_1, \ldots, \mathcal{G}_{i-1}$, carries the largest determinant weight. To this end set $m = \sum_{j<i} r_j$, and put $B = \sum_{j<i} r_j \gamma_j$. By Proposition proposition-determinant-weights-tensor-operations, the largest determinant weight of a constituent of $\bigwedge^{m+1} \mathcal{G}$ is $B + \gamma_i$. Indeed, it is obtained by taking $n_j = r_j$ for every $j < i$ and $n_i = 1$, and every other choice gives a smaller sum, because the γ_j are strictly decreasing. Moreover, exterior powers of a real sheaf are real. Indeed, their eigenvalues are the products of eigenvalues with distinct indices, and complex conjugation permutes this multiset. Hence Proposition 14.25 applies to $\bigwedge^{m+1} \mathcal{G}$.

The eigenvalues of $\mathcal{G}_i$ appear in $\bigwedge^{m+1} \mathcal{G}$ not by themselves but multiplied by the eigenvalues of the determinants of the groups of larger weight, and those eigenvalues are known. To make this precise, let x be a closed point of the curve and α an eigenvalue of Frob_x on $(\mathcal{G}_i)_{\bar{x}}$. The decomposition of the exterior power of a direct sum shows that $\bigwedge^{m+1} \mathcal{G}$ contains the summand

$$(\det \mathcal{G}_1 \otimes \cdots \otimes \det \mathcal{G}_{i-1}) \otimes \mathcal{G}_i.$$

If β_j is the eigenvalue of Frob_x on $(\det \mathcal{G}_j)_{\bar{x}}$, then the product $\alpha\beta_1 \cdots \beta_{i-1}$ is an eigenvalue of Frob_x on the stalk of this summand at $\bar{x}$. Since $\det \mathcal{G}_j$ is pure of weight $r_j\gamma_j$, we have $|\iota(\beta_1 \cdots \beta_{i-1})|^2 = q_x^B$. Now Proposition 14.25 gives

$$|\iota(\alpha)|^2 q_x^B \le q_x^{B+\gamma_i}, \qquad |\iota(\alpha)|^2 \le q_x^{\gamma_i}.$$

For the reverse bound, note that the dual $\mathcal{G}^\vee$ is real as well, with determinant weights $-\gamma_t > \cdots > -\gamma_1$. Applying the same argument to the summand $\mathcal{G}_i^\vee$, we get $|\iota(\alpha^{-1})|^2 \le q_x^{-\gamma_i}$, and hence $|\iota(\alpha)|^2 = q_x^{\gamma_i}$. Since every irreducible constituent of $\mathcal{G}$ is a summand of exactly one $\mathcal{G}_i$, this proves the theorem on an affine curve.

On an arbitrary curve we restrict to an affine neighbourhood of a given closed point x, which suffices because purity is a condition at each closed point. It remains to check that the restriction changes neither the constituents nor their determinant weights. The neighbourhood is a nonempty open subscheme of a smooth geometrically integral curve, so it is geometrically connected, and restriction to it preserves irreducibility. Indeed, a connected finite étale cover of a normal connected curve is integral, and hence remains connected after restriction to a dense open. So the maps of geometric and of arithmetic fundamental groups are surjective, and since restriction preserves the degree map, the map of Weil groups is surjective too. Hence a subspace of the fibre is stable on the open if and only if it is stable on the curve.

The restriction of a real sheaf is real, and by Corollary 14.6 the determinant keeps its weight. The affine case therefore applies at x, and since x was arbitrary, we obtain purity on the whole curve. Finally, the sheaf is mixed because a composition series is a finite filtration with pure quotients. ∎

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